Cristiano Capone
National Center for Radiation Protection and Computational Physics
Istituto Superiore di Sanità, 00161 Rome, Italy
cristiano.capone@iss.it · July 27, 2026

The Editors, *Nature Communications*

Dear Editors,

We are pleased to submit our manuscript, "From read-out geometry to in-silico stimulation: a distributed functional-connectivity signature of Alzheimer's disease," by Cristiano Capone, Enza Cece, Andrea Ciardiello, Guido Gigante, Evaristo Cisbani and Maurizio Mattia, for consideration as an Article in *Nature Communications*.

Network-level neuromodulation is most effective when a single, well-defined site can be engaged, yet Alzheimer's disease is widely regarded as a distributed network disorder. Whether its connectivity signature can be reduced to a focal, stimulable target has never been addressed causally, because doing so requires an individualised generative model of brain dynamics that can be perturbed *in silico*. We fit subject-specific, cross-subject-identifiable reservoir-computing models that reconstruct each individual's lagged functional connectivity and use them as a testbed for such counterfactual interventions.

The central advance of our work is a clean conceptual distinction between two ways of intervening on such a model: perturbing the connectivity *graph* (the fitted read-out kernel that defines the network's interactions) and perturbing *activity* (a focal oscillatory drive injected at a node). We show these are fundamentally different interventions. Restoring control-like dynamics by changing the graph is *intrinsically distributed*: the required correction is a coordinated, multi-site change of the connectivity kernel that cannot be reproduced by editing any single node: a single-node graph edit is inert. A focal activity perturbation, by contrast, *can* be effective from a single, physically realisable site.

Crucially, the site at which focal activity stimulation is optimal is *not* the site at which the graph is most altered. The "graph" sites (those with the largest change in the connectivity kernel, predominantly weakly-connected subcortical nodes) and the "activity" sites (those optimal for focal stimulation, predominantly well-connected cortical nodes) are largely disjoint. This dissociation is governed by the connectivity properties of the graph itself: the most graph-perturbed nodes are the most weakly coupled, and therefore poor actuators, whereas the nodes that provide therapeutic leverage are those best positioned to redistribute a corrective perturbation network-wide. The optimal target must therefore be chosen by its effect on the network's dynamics, not by where the change in the graph concentrates. We further show that a real-time closed-loop controller attains comparable efficacy at substantially lower dose using only causally available information, the in-silico analogue of adaptive neuromodulation.

These results reframe a central question in targeted neuromodulation, "where should we stimulate?", as a question about the interaction between intervention modality and network connectivity, answerable only with an individualised, perturbable model. We believe this distinction between graph and activity perturbation, grounded in the graph's connectivity properties, will interest the broad readership of *Nature Communications* across network neuroscience, computational modelling and neuromodulation, well beyond Alzheimer's disease.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. The data derive from the Alzheimer's Disease Neuroimaging Initiative and are used under its Data Use Agreement. We declare no competing interests, and would gladly suggest suitable referees on request. We thank you for considering our work.

Sincerely,
Cristiano Capone, on behalf of all authors